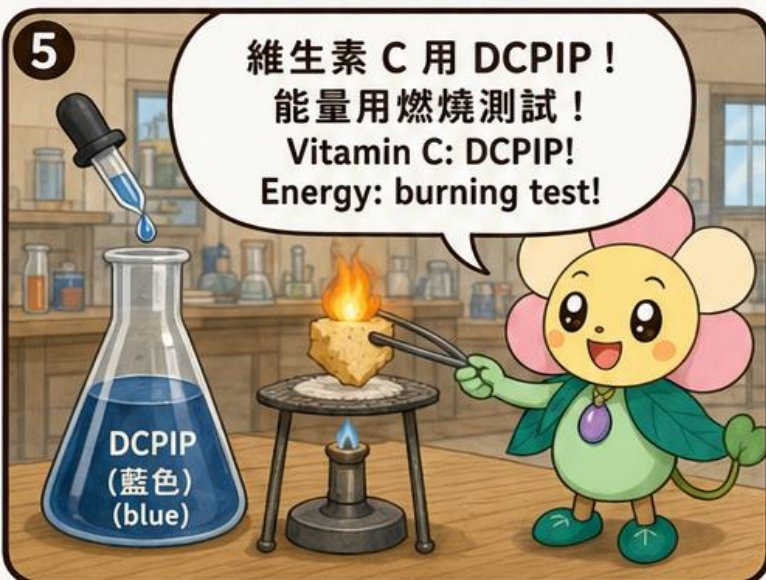
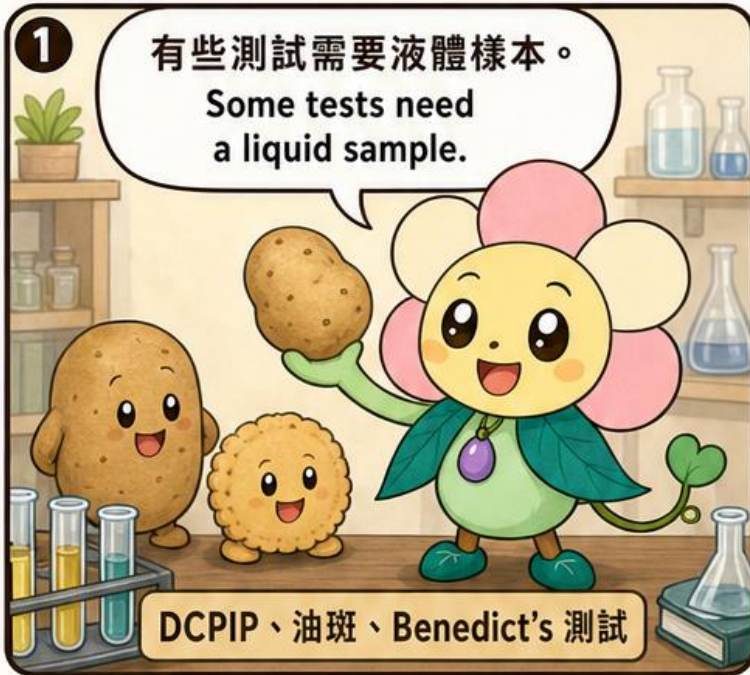


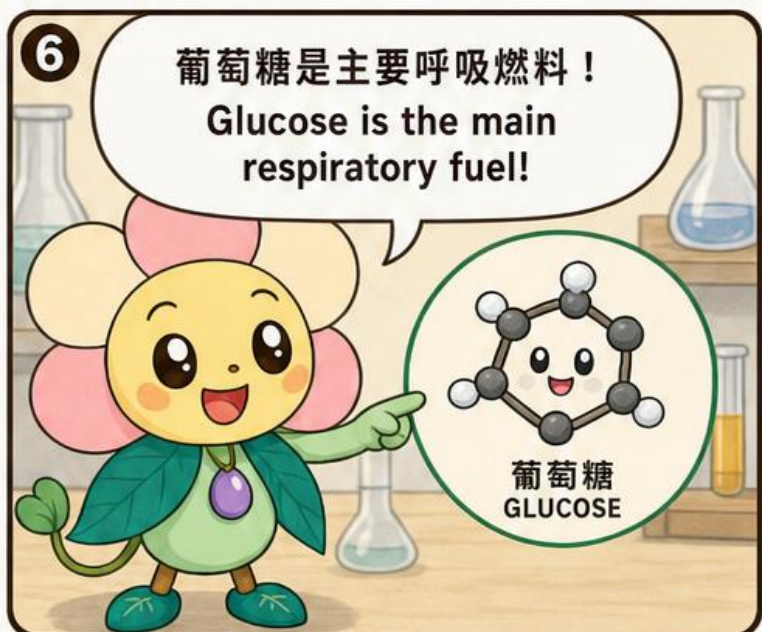
《花瓣仔的食物偵探實驗室》 Petalino and the Food Detective Lab



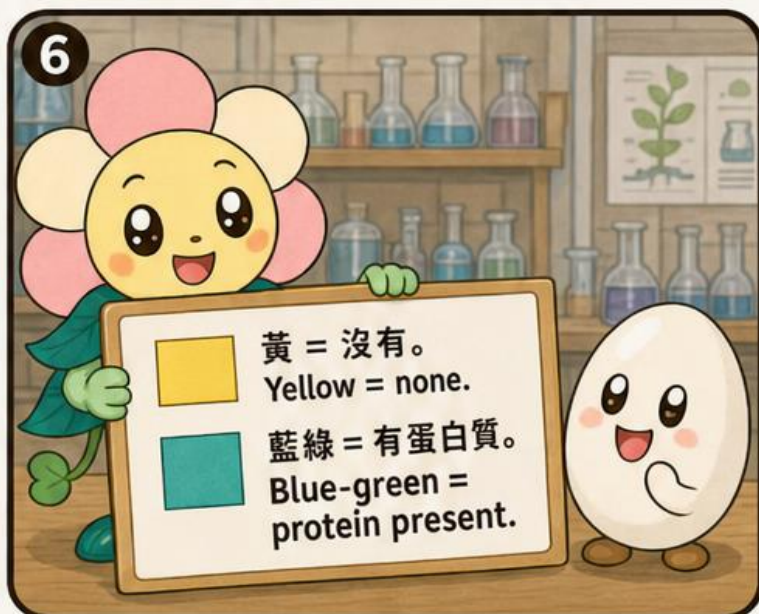


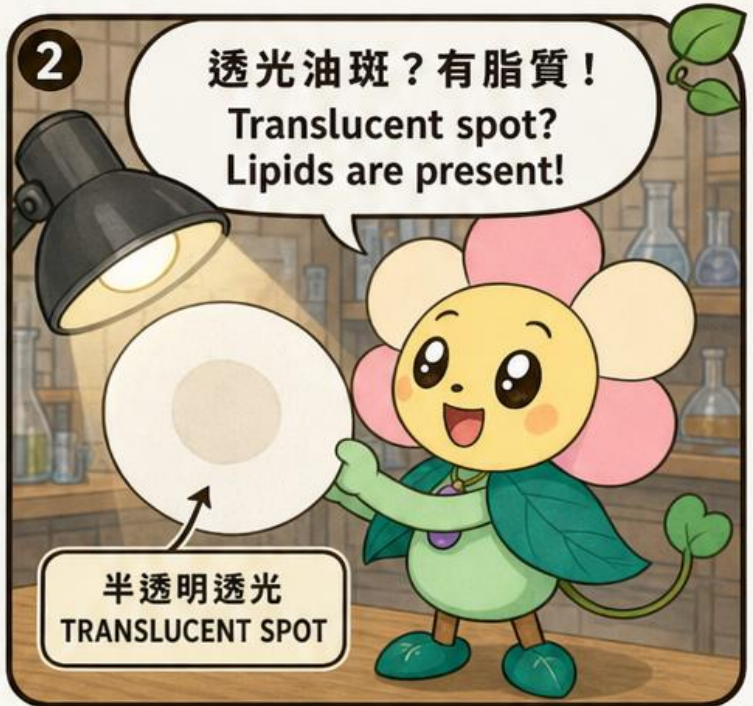
切碎 → 研磨 → 過濾 → 提取液
Cut → grind → filter → extract

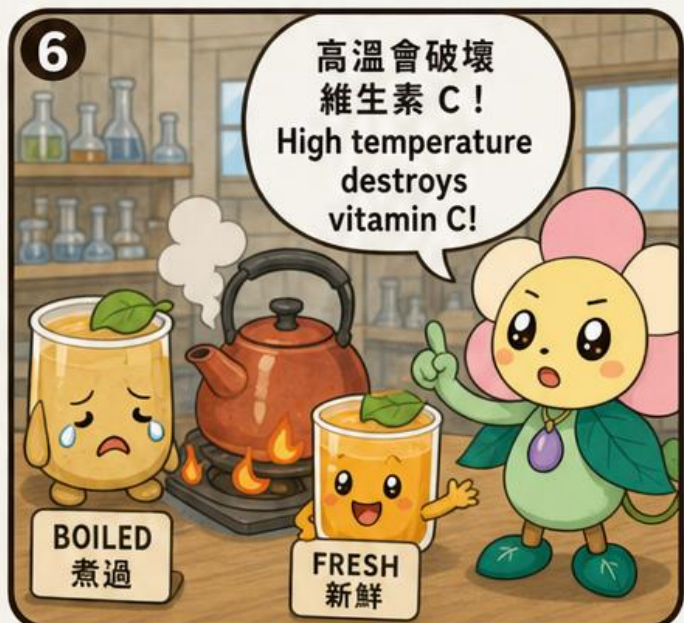
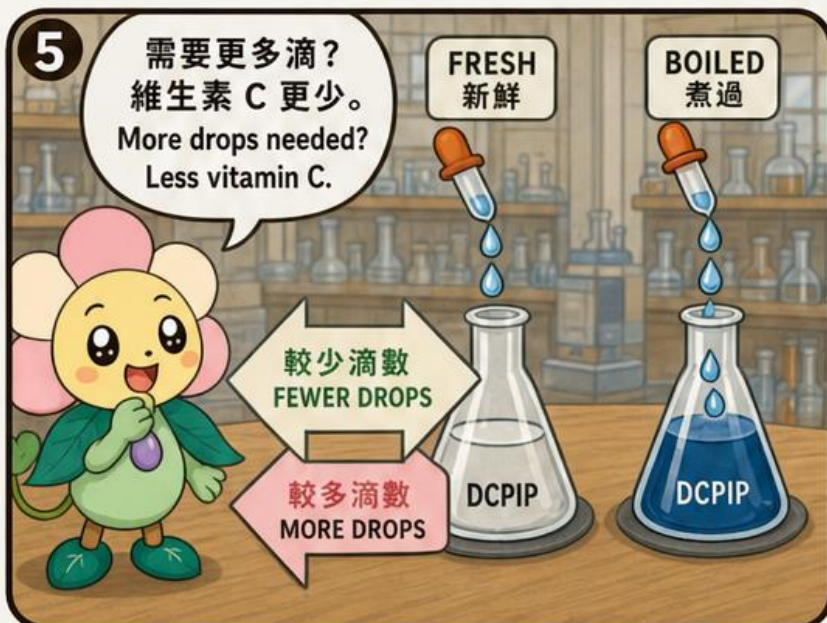
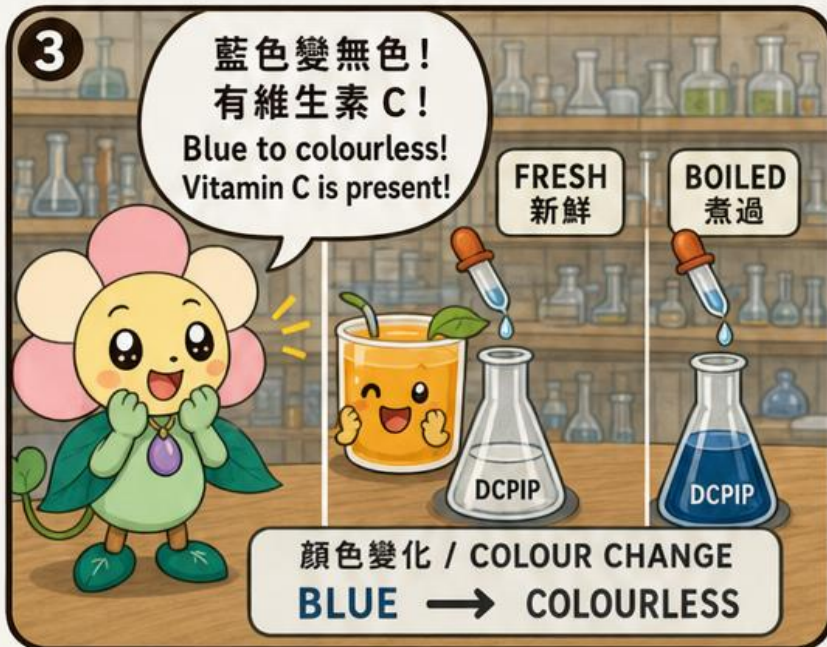




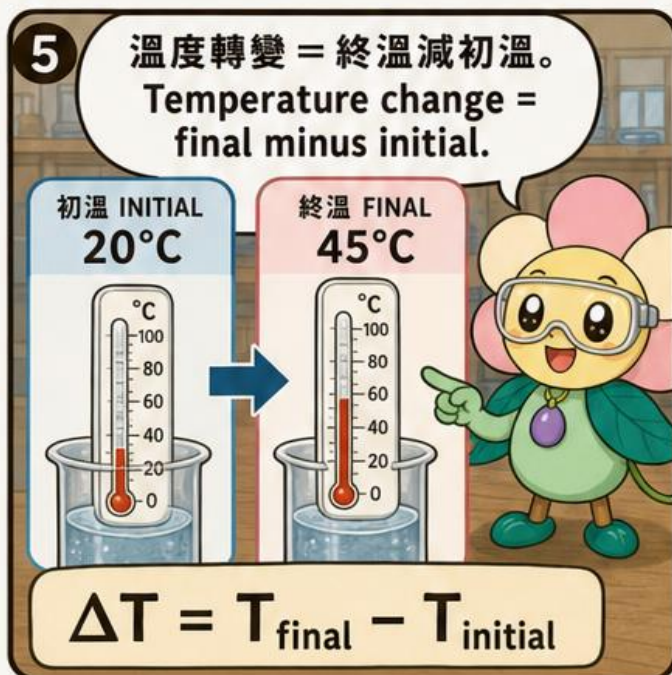








較少滴數 = 較高濃度 / Fewer drops = higher concentration



1

**LIPIDS
×2**

脂質必須
乘二！
Lipids must
be doubled!

2

每克脂質能量是碳水或
蛋白質的兩倍！
Energy per gram of lipid is
double that of carbohydrate
or protein.

1 g 脂質
1 g lipid

1 g 碳水 / 蛋白質
1 g carb/protein

3

計算前，脂質先×2！
Multiply lipids by 2
before you add them up!

4

誤差一：熱量散失到周圍。
Error 1: heat is lost
to the surroundings.

不是所有能量都用來加熱水
Not all energy heats the water

5

誤差二：不完全燃燒。
Error 2: incomplete
combustion.

不是所有化學能都變成熱
Not all chemical energy
becomes heat

6

所以量到的能量
低於真正能量！
So the measured energy
is less than the actual energy!

低估
UNDERESTIMATION

散失熱 + 不完全燃燒 → 低估
Heat loss + incomplete combustion → underestimation

《花瓣仔的食物偵探實驗室》 Petalino and the Food Detective Lab

1

葡萄糖試紙：綠→棕。
磷液：棕→藍黑。
Glucose paper: green to brown.
Iodine: brown to blue-black.



2

Albustix：黃→藍綠！
Albustix: yellow to blue-green!



3

油斑透光，
加酒精後消失！
Translucent grease spot,
then it disappears
in alcohol!



4

DCPIP：藍→無色。
滴數越少，濃度越高！
DCPIP: blue to colourless.
Fewer drops, higher
concentration!



5

燃燒測試：
溫差越大能量越多。
脂質×2！
Burning test:
greater ΔT means
more energy.
Lipids × 2!

低估誤差
UNDERESTIMATION



6

現在我們可以像偵探一樣
辨認食物成分！
Now we can identify food
substances like detectives!



結案！ CASE CLOSED!



磷粉只在植物來源
Phytate only in plants



無斑 = 無脂質
No spot = no lipids



更多DCPIP滴 = 更少維生素C
More DCPIP drops = less vitamin C



燃燒誤差會低估
Burning error underestimates